

MD. BAYZID HOSSAIN

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Data Engineer

EDUCATION

- International University of Business Agriculture and Technology**

Bachelor of Science in Computer Science and Engineering

May 2022 - April 2026

CGPA: 3.44 in the scale of 4.00

SKILLS AND AREA

Programming Languages	Python, SQL, C, C++, Java
Cloud & Warehouses:	AWS, Azure, Snowflake, Redshift, MS Fabric, Synapse Analytics
ETL & Orchestration	Airflow, DBT, AWS Glue, Azure Data Factory
Stream & Batch Processing	Kafka, PySpark, Spark, Databricks
Data Modeling	Star Schema, Medallion Architecture, Dimensional Modeling
Databases & Formats	MySQL, PostgreSQL, Delta Lake, Parquet, JSON
DevOps & Tools	Docker, Kubernetes, Git/GitHub, Jupyter Notebook

TRAINING AND CERTIFICATION

- Edge Training project:** [Database\(MySQL/Oracle/SQL Server\)](#)

Nov '24 - May '25

PROJECTS

Spotify Streaming Data Pipeline [Kafka, Snowflake, DBT, Airflow, MinIO, Docker]

[GitHub](#)

- Developed streaming platform ingesting 3,600+ user events/hour (plays, skips, playlists) via Kafka, storing time-partitioned data in MinIO with hourly Airflow-to-Snowflake loads
- Applied DBT transformations to generate 7+ Gold-layer models for user engagement metrics (DAU, retention, skip rates) in Snowflake
- Deployed containerized architecture with Docker Compose for Kafka, Zookeeper, Airflow, and MinIO

Snowflake Cricket Analytics Pipeline [Snowflake, Star Schema, Window Functions]

[GitHub](#)

- Engineered enterprise data warehouse processing 2,411 cricket match JSON files through 4-layer architecture (Landing → Raw → Curated → Enriched) with star schema design (6 dimensions, 2 fact tables)
- Utilized `LATERAL FLATTEN` for nested JSON parsing, window functions for analytical calculations, and transient tables handling 500K+ ball-by-ball delivery records
- Implemented Type 1 SCD with surrogate keys and foreign key constraints, enabling complex analytics across match outcomes, player statistics, and venue performance

AWS Serverless Flight Delay ETL Pipeline [AWS Glue, Step Functions, Redshift, S3, EventBridge]

[GitHub](#)

- Architected event-driven ETL pipeline with AWS Glue and PySpark, orchestrated via Step Functions with automated crawler polling, processing 2,400+ daily flight records
- Enriched data through complex joins with airport dimensions, loading denormalized Redshift tables achieving 10–50x faster BI query performance
- Configured S3 EventBridge triggers and SNS monitoring for fully automated, zero-touch pipeline execution

Azure Healthcare Streaming Pipeline [Azure Event Hubs, Databricks, Delta Lake, Synapse Analytics, Power BI]

[GitHub](#)

- Built real-time streaming pipeline with Event Hubs ingesting 3,600+ patient events/hour through Medallion Architecture (Bronze → Silver → Gold) using PySpark on Databricks
- Devised data quality framework with automated corrections for invalid records (future timestamps, age outliers) and dynamic schema evolution handling
- Executed SCD Type 2 in Delta Lake tracking patient history with effective dating, exposed via Synapse SQL for Power BI analytics